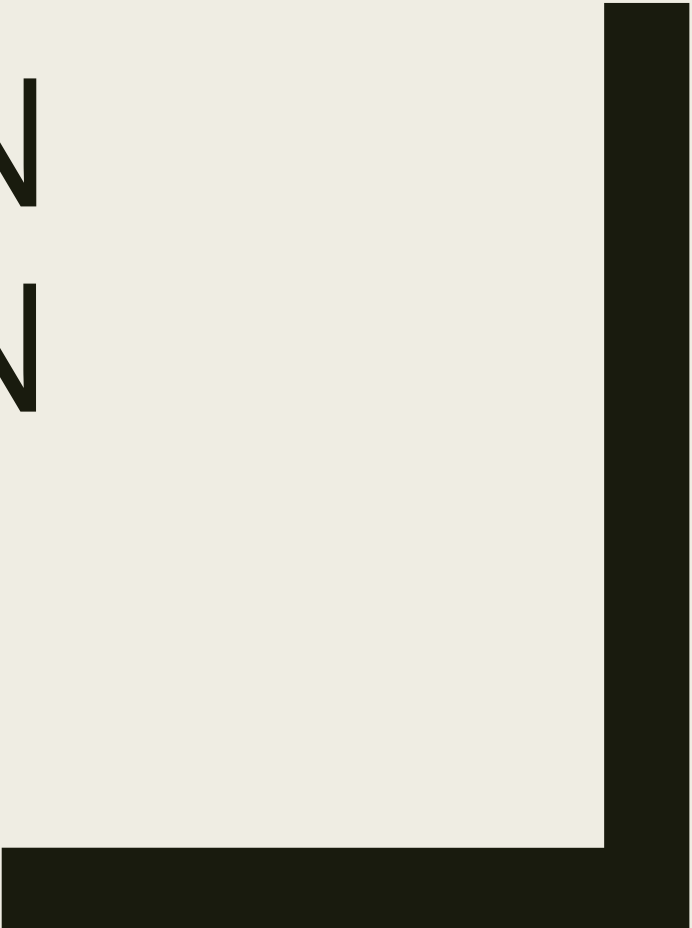




DIMENSION REDUCTION



Outline

- Motivation for Dimension Reduction
- Types and Relationship to other Techniques
- Exploratory Factor Analysis
- Principal Components Analysis

Dimension Reduction...

- ... is used in data science and machine learning to reduce the number of features while preserving as much relevant information as possible

Purpose

- Clarify Structure
 - *Simplify interpretation of data by grouping correlated variables into components*
 - *Reveal latent factors*
- Mitigate the curse of dimensionality. Parsimony is a desirable property for models
 - *Predictions from simple (versus complex models) are more stable across samples*
 - *Simple models are easier to interpret and communicate to stakeholders.*
 - *Certain models cannot be used if number of features exceeds sample size*
 - *Reduce multicollinearity*
 - *Reduce computational requirements*

Uses

- In survey research, reduce a large number of survey items into a smaller set of meaningful factors
- In domains where high-dimensional data is common (e.g., images, text, genomics), reduce dimensions while preserving information
- Help visualize high-dimensional data by reducing it to two or three dimensions.

Suitability

- Dimension Reduction should be employed when
 - *There are many features relative to observations*
 - *Features are correlated*
 - *Computational efficiency is important*
 - *Features represent underlying latent factors*
 - *Data needs to be visualized*
- On the other hand, dimension reduction should not be used when
 - *All features are equally important*
 - *Interpretability of original features is critical*
 - *Data is already low-dimensional*
 - *Sample size is small*

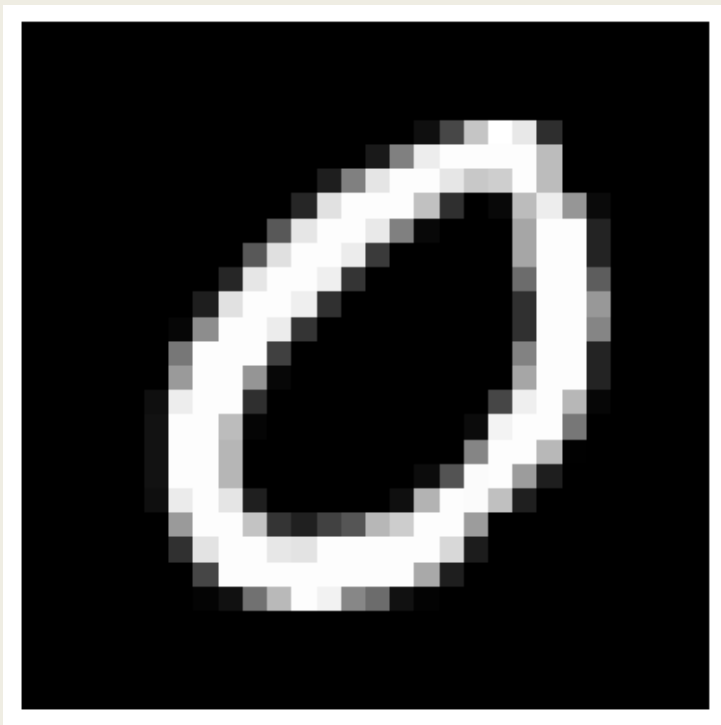
Dimension Reduction vs. Clustering

- Like clustering, dimension reduction techniques are unsupervised learning approaches
- In contrast to clustering, dimension reduction seeks to group variables rather than observations.

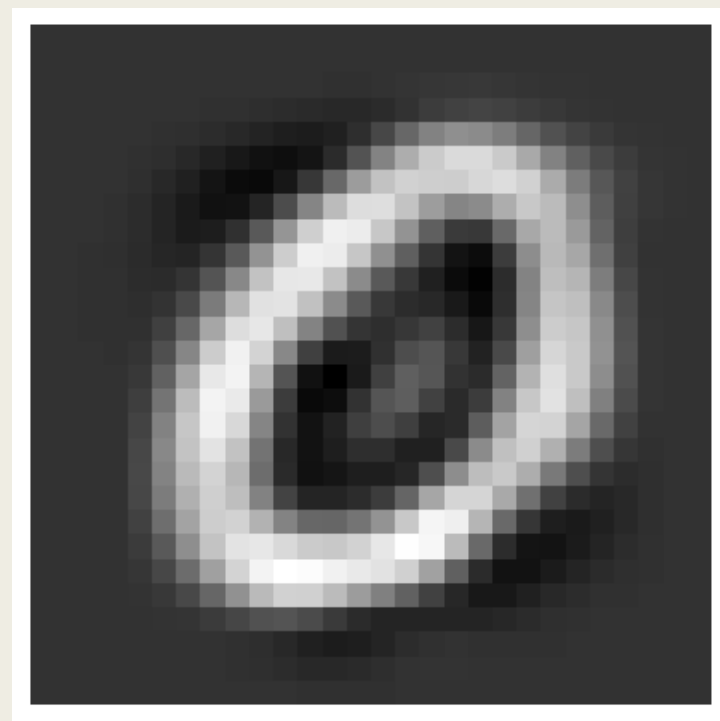
Metaphors

- Dimension reduction works by grouping similar variables together.
 - *To better understand the process of dimension reduction, think of spinning a test tube containing muddy water in a centrifuge. Similar particles are grouped together in layers, each layer represents a factor or component of the original data.*
- Dimension reduction involves loss of information
 - *Representing a large number of variables using a small number of factors or components often results in loss of information. However, this is okay so long as the information loss is not substantial. To better understand this idea, think of .jpg or .gif image files used on websites. These are compact image files that look just like the original .bmp or .tiff image files they are derived from. It is only at high resolution that the pixelation of a .gif or .jpg is apparent.*

Original Data



70% of Original Data



Approaches based on Purpose

- Determine underlying themes
 - *Identify underlying dimensions or factors that explain the correlations among a set of variables. This is frequently used in the social sciences to identify latent variables or constructs underlying measured variables.*
 - *Technique: Factor Analysis*
- Represent data with fewer components
 - *To identify a set of uncorrelated variables to replace the original set of correlated variables. This may be done to use the variables in a supervised learning technique that is sensitive to multicollinearity.*
 - *Technique: Principal Components Analysis*
- Note: The distinction is not as cut and dry as indicated by the above classification. Factor analysis may also be used to reduce data while Principal Components Analysis may shed light on underlying meaning

FACTOR ANALYSIS

Purpose of Factor Analysis

- A fundamental assumption of factor analysis is that measured data represent deeper latent factors.
- These factors cannot be directly measured but influence observed variables. E.g., attitude toward the brand is a latent factor or construct measured by a set of survey items.
- It aims to find deeper latent factors underlying observed data or to confirm relationships between observed data and latent factors.
 - *For e.g., if the questions included in the survey represent the constructs they were designed to measure.*

Expected Structure

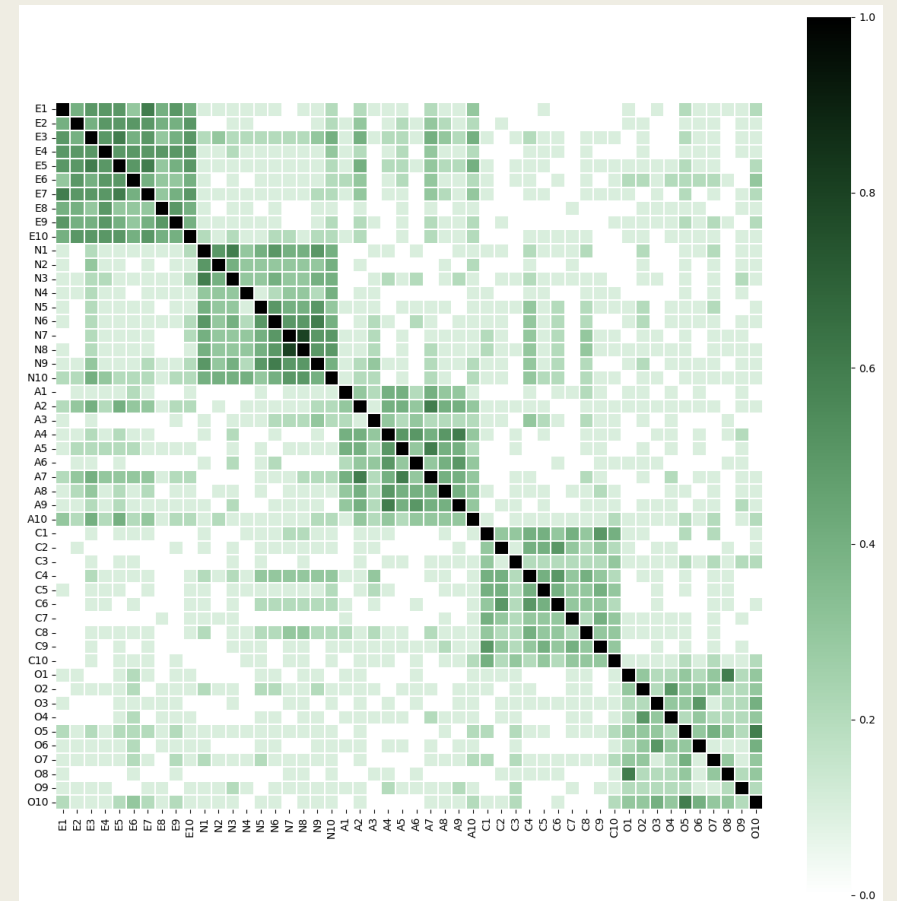
- Unlike many supervised learning techniques which generate a unique solution, factor analysis generates many good solutions.
- Factor analysis assumes the existence of a theoretical mapping between the factor and variables.
- Thus, before conducting a factor analysis, it is important to formulate an expectation of the latent factors underlying the data and their relationship with the observed variables.

Steps in Factor Analysis

- Assess suitability for Factor Analysis
 - *Data must contain a pattern of relationships among variables, sample must be large enough relative to number of features*
- Prepare Data
 - *Missing data issues must be addressed, irrelevant features removed, and features for factor analysis converted to the same scale.*
- Determine number of factors
 - *Factor analysis relies on strong theory. Metrics such as eigen-value threshold, variance explained, and parallel analysis should be interpreted in the context of the theory.*
- Mapping Features to Factors
 - *Since interpretability is key, results of factor analysis are often rotated in search of a meaningful solution.*
- Interpret Factors

Suitability for Factor Analysis

- Pattern of correlations
- Bartlett's Test of Sphericity
 - *Compares correlation matrix to identity matrix*
- KMO Measure of Sampling Adequacy
 - *KMO MSA >> 0.5*
- Number of features << Number of observations
- Large enough sample. $n > 100$

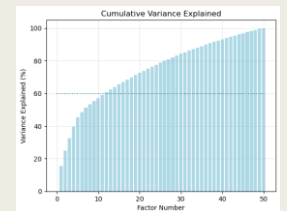
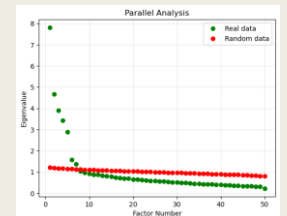
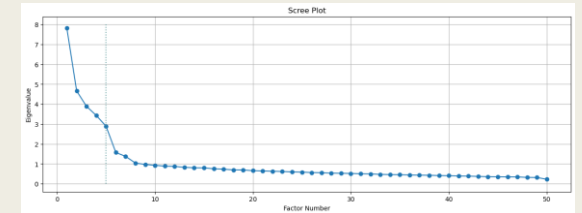


Data Preparation

- Missing values: Impute missing values to avoid listwise deletion, the default
- Standardize Variables: Factor analysis is sensitive to measurement scale
- Categorical variables: Unless, absolutely necessary categorical variables should not be included as factor analysis algorithms work best when all variables are of the same type. If necessary, dummy code categorical variables before including in analysis.
- Outliers: Extreme observations may bias factor structure

Determine Number of Factors

- It is critical to have an a priori expectation of the number of factors.
- Following data-driven methods must be used to corroborate an a priori solution or select a solution in the neighborhood
 - *Eigen-value*
 - Select all factors with eigen value greater than 1
 - *Elbow in Scree Plot*
 - Line graph of eigen values for each factor. Ideal number of factors is indicated by a sudden change in the line graph or what is known as the *elbow*.
 - *Parallel Analysis*
 - Compare eigen values in data to those obtained from simulating random data.
 - *Total Variance Explained*
 - To ensure that the factors represents the original variables sufficiently well, the total variance explained by factors should be greater than 70%.
 - *Extracted Communalities*
 - Communality reflects the amount of variance in a variable that can be explained by the factors.

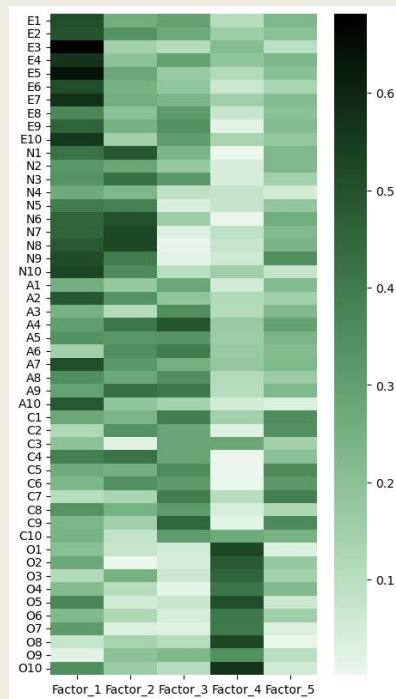


Mapping Features to Factors

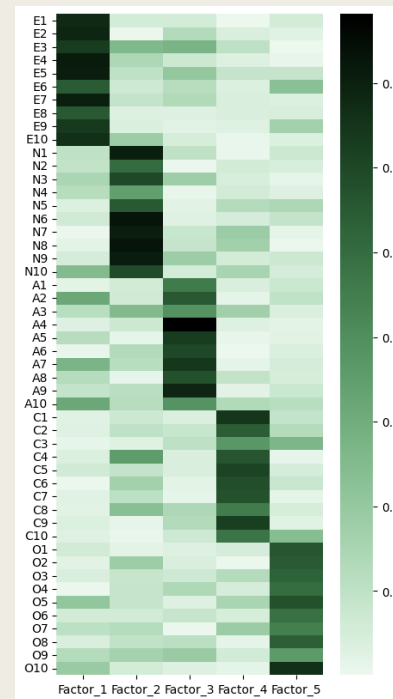
- Each variable is represented as a linear combination of factors
- An ideal factor solution is where each variable is expected to load on (i.e., related to) only one factor. Such a result is easy to interpret.
- In practice, each variable may load on many factors. This may still be acceptable so long as the loading on one factor is large and on all other factors is small.
- When the pattern of loadings does not show a clear preference of a variable for a factor, rotating the axes may help generate a clear mapping. There are two broad types of axes rotation
 - *Orthogonal: Axes are rotated while constraining them to be at right angles. E.g., varimax, quartimax, equimax,*
 - *Oblique: Axes are allowed to have any angle between them. E.g., oblimin, promax*

Mapping Features to Factors

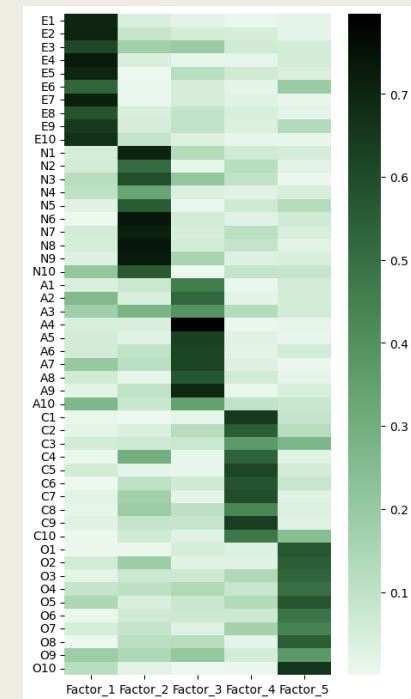
Unrotated



Varimax



Oblimin



Interpretation and Representation

- Review pattern of factor loadings from the rotated matrix to interpret the factor analysis solution
- The meaning of each factor is derived from the variables loading on it.
- Review the variables to describe each factor.
- Combine features into factors by computing
 - *arithmetic mean of the items loading on it*
 - *weighted mean of items loading on it where weights are derived from factor loadings*

PRINCIPAL COMPONENTS ANALYSIS

Principal Components Analysis

Principal Component Analysis (PCA) is a powerful linear dimensionality reduction technique. It reduces dimensionality of data while preserving as much of the original information as possible

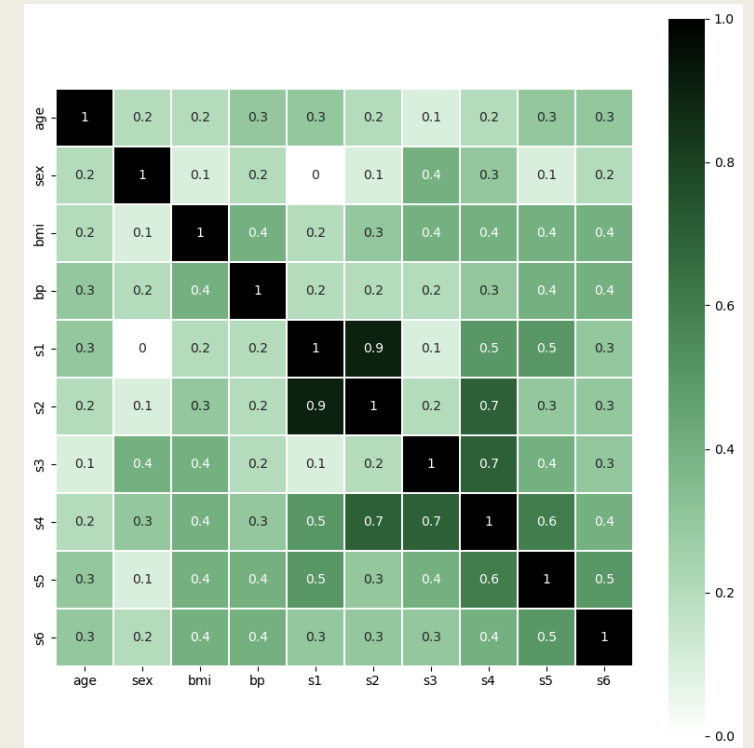
- Used to reduce the dimensionality of the data by representing a large number of variables with a fewer number of components. Similar variables get grouped into the same component while dissimilar variables are placed in different components.
- Like factor analysis, it reduces the data based on similarity (typically inferred from correlation).
- It is used when the primary concern is to determine the minimum number of components that will account for maximum variance in the data for use in subsequent machine learning tasks.
- Not concerned with uncovering meaning of components
- This reduced number of components can be used for further analysis instead of the original set of variables.
- Is computationally efficient
- Most widely used dimension reduction technique

Steps in PCA

- Assess suitability for PCA
 - *Data must contain a pattern of relationships among variables, sample must be large enough relative to number of features*
- Prepare Data
 - *Missing data issues must be addressed, irrelevant features removed, features converted to the same scale, split data, and drop outcome variable*
- Determine number of components
 - *Examine eigen value, variance explained and scree plot*
- Derive components
 - *Apply component structure to test set and extract components*

Suitability for PCA

- Since Principal Components Analysis attempts to group similar variables, a basic requirement is that at least a few variables be related to each other.
- Correlation Matrix
 - *Looking for a pattern of correlations across features*



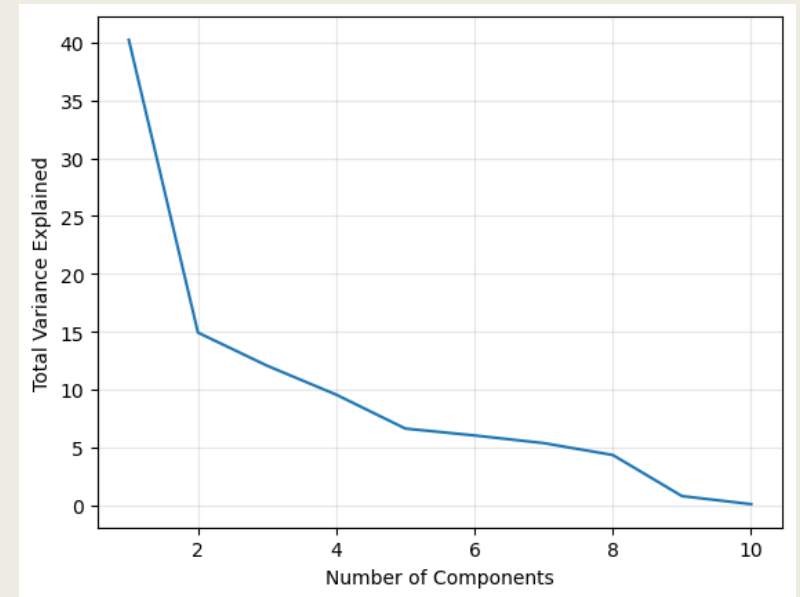
Prepare Data

- Missing Values: Principal Components Analysis uses data on all variables. A missing observation on one variable will cause the entire row of data to be ignored for analysis. Therefore, it is important to impute missing values.
- Standardize: It is important to standardize variables to ensure they receive the same weight in analysis.
- Split Data: Generate PCA schema using train sample and apply it to test sample.
- Drop irrelevant variables: PCA is an unsupervised learning technique, so it only operates on features, so remove outcome variable. Also, remove irrelevant features.

Determine number of components

A dataset with p variables will generate p components. The goal is to pick out the top few components that capture most of the variance in the original data. This is done based on the following criteria.

- Total Variance Explained
 - *To ensure that the components represents the original variables sufficiently well, the total variance explained by components should be greater than 70%.*
- Scree Plot
 - *Line graph of eigen values for each component. Ideal number of factors is indicated by a sudden change in the line graph or what is known as the elbow.*
- Eigen-value
 - *Select all components with eigen value greater than 1*



Derive Components

- In order to use the components, for downstream analysis,
 - *apply the component structure to the test set.*
 - *extract the components*
 - *combine components with other variables in the original dataset*

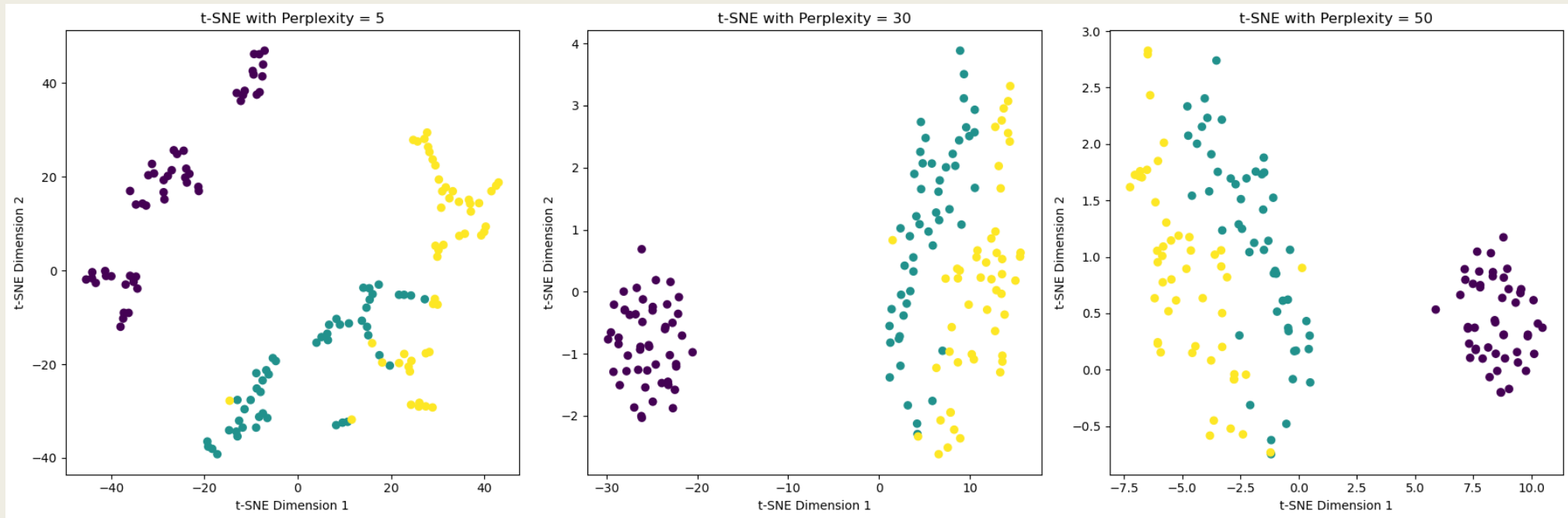
MANIFOLD LEARNING

t-SNE

t-Distributed Stochastic Neighbor Embedding

- PCA does not work well with non-linear relationships
- Can capture and represent complex non-linear relationships within high dimensional data
- t-SNE constructs probability distributions over pairs of points in both the high-dimensional and low-dimensional spaces, then minimizes the difference between these distributions.
- As a result, points that are close in the original space tend to remain close in the reduced space, while maintaining a degree of separation between dissimilar points.

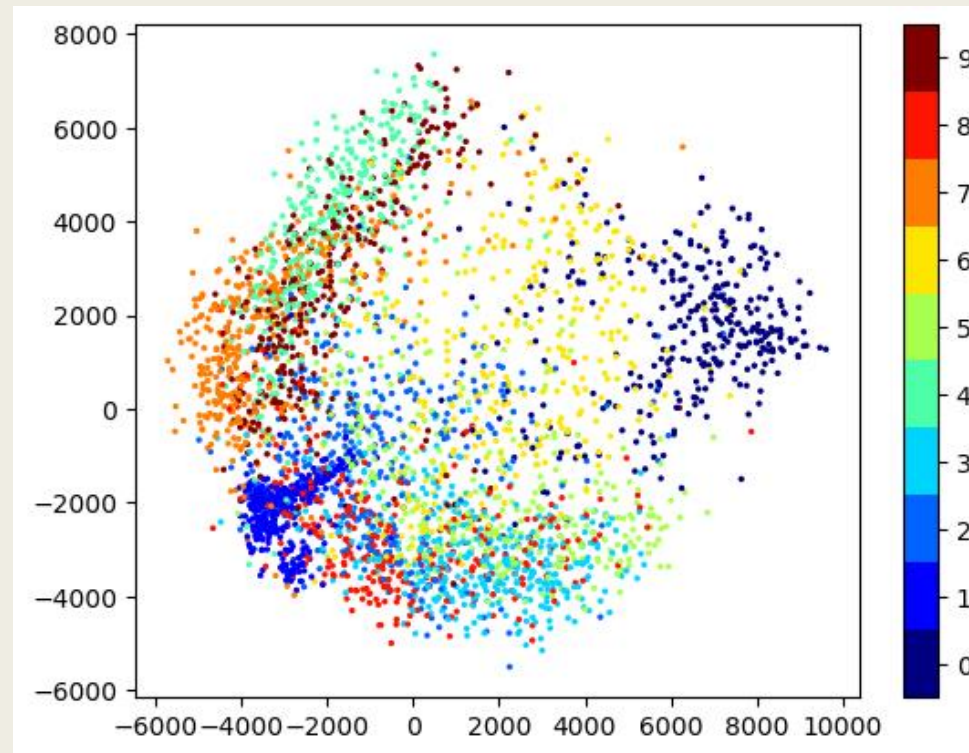
t-SNE



Manifold Learning

- A class of unsupervised estimators that seek to describe datasets as low-dimensional manifolds embedded in high-dimensional spaces.
- There are a number of manifold methods, including multidimensional scaling (MDS), locally linear embedding (LLE), and isometric mapping (Isomap).
- We will limit our focus to isometric mapping.

Isometric Mapping



Conclusion

- In this module, we
 - *discussed the motivation for conducting dimension reduction*
 - *examine factor analysis for deriving latent factors from a set of features*
 - *discussed the use of PCA for reducing a large number of features to a smaller set of components*
 - *discussed manifold learning to aid with visualization*